SMART FACIAL SCREENING FOR DS DETECTION

Automated AI-assisted screening - not a diagnosis

Patient Information

Name: ma

Date: 2026-03-04 13:29:42

Child Age: 13

Mother Age: 36

Father Age: 43

Parents Relation: distant_relative

Living Area: urban_area

Previous Pregnancies: 0

Scre

Probability: 85%

Confidence: 85%

Risk Level: High

High Confidence: No

Facial Features (AI analysis)

Facial Symmetry: 0.86 Eye Spacing: 0.61 Nasal Bridge: 0.70 Ear Position: 0.56

AI Explanation

{
"title": "**High Risk for Down Syndrome Traits**",
"explanation": "The patient has a high probability of Down syndrome traits, as indicated by the high risk level. The Grad-CAM heatmap shows significant activation in the eyes, nose, and mouth regions. The eyes have an activation score of 0.628, indicating a 62.8% chance of involvement. The nose has an activation score of 0.606, suggesting a 60.6% chance, and the mouth has an activation score of 0.433, indicating a 43.3% chance. These regions are crucial for facial recognition and speech development, which are critical for future health challenges. The risk factors include facial symmetry, eye spacing, nasal bridge, and ear position, which are all important for facial recognition and speech development. The affected regions are likely to present health challenges such as speech delay, feeding difficulty, and slower milestones. The patient should monitor these health issues closely. The patient should eat a balanced diet rich in vitamins and minerals, and engage in physical activities such as yoga and swimming to support overall health and development.",

"affected_regions": ["eyes", "nose", "mouth"],
"food_recommendations": [Risk overview: High likelihood (85%) based on Grad-CAM facial pattern analysis. Affected facial regions: - eyes: score 0.628, severity high. Peri-orbital spacing/alignment patterns were highlighted in this region. - nose: score 0.606, severity high. Mid-face and nasal-bridge structure patterns were highlighted in this region. - mouth: score 0.433, severity moderate. Lower mid-face/oral contour patterns were highlighted in this region. Risk factors (Grad-CAM): eyes: 0.628 (high, 27.4%); nose: 0.606 (high, 26.4%); mouth: 0.433 (moderate, 18.9%); forehead: 0.337 (low, 14.7%). Risk factors (facial features): facial_symmetry=0.86 (high); eye_spacing=0.61 (high); nasal_bridge=0.7 (high); ear_position=0.56 (moderate). Future health issues to monitor: - Speech and language delay [high]: Consider early speech and language support. - Low muscle tone [high]: Track posture, motor milestones, and physiotherapy needs. - Feeding or swallowing difficulty [high]: Monitor feeding tolerance and growth trend. - Slower developmental milestones [high]: Use periodic developmental screening follow-up. - Oral-motor articulation concern [high]: Watch speech clarity and oral-motor control. - Breathing/sleep quality observation [high]: Track persistent nasal obstruction or sleep issues. Support plan: Food - Omega-3 rich foods, leafy greens, fruits. Exercise - Eye focus drills, gentle neck stretches, breathing, supervised physiotherapy. This is a screening result, not a final diagnosis. Confirm clinically with a pediatric specialist.

Recommendations

Food:
- Omega-3 rich foods
- leafy greens
- fruits

Exerc
- Eye
- gent
- brea
- supe

Detailed Result Fields

Risk Factor Explanation: eyes: score 0.628, severity high, contribution 27.4%; nose: score 0.606, severity high, contribution 26.4%; mouth: score 0.433, severity moderate, contribution 18.9%; forehead: score 0.337, severity low, contribution 14.7%; facial_symmetry: value 0.86, severity high; eye_spacing: value 0.61, severity high; nasal_bridge: value 0.7, severity high; ear_position: value 0.56, severity moderate

Affect
in this
region

This report is for screening purposes only and is not a diagnosis.